

Conditional Probability and Bayes' Theorem

Complete detailed notes with formulas, examples, and exam-ready shortcuts

1. What is Probability?

Probability means the chance or likelihood that an event will happen. It is used whenever the result is uncertain, such as tossing a coin, rolling a die, drawing a card, or predicting whether a customer will buy a product.

$$P(E) = \text{Number of favourable outcomes} / \text{Total number of outcomes} = n(E) / n(S)$$

The value of probability always lies between 0 and 1. If $P(E) = 0$, the event is impossible. If $P(E) = 1$, the event is certain. If the value is closer to 1, the event is more likely to happen.

Example: When a fair die is rolled, $S = \{1, 2, 3, 4, 5, 6\}$. The probability of getting an even number is $P(E) = 3/6 = 1/2$ because the favourable outcomes are $\{2, 4, 6\}$.

2. Basic Terms

Term	Meaning	Example
Experiment	An action with an uncertain outcome.	Rolling a die, tossing a coin, selecting a card.
Sample Space (S)	Set of all possible outcomes.	For a die: $\{1,2,3,4,5,6\}$.
Event	A subset of the sample space.	Getting an even number: $\{2,4,6\}$.
Favourable Outcomes	Outcomes that satisfy the event.	For even number on die: 2, 4, 6.

3. Important Types of Events

3.1 Complementary Events

If E is an event, then the complement of E means 'not E'. It is written as E' or E^c .

$$P(E) + P(E') = 1 \text{ therefore } P(E') = 1 - P(E)$$

Example: If probability of rain is 0.30, then probability of no rain is $1 - 0.30 = 0.70$.

3.2 Mutually Exclusive Events

Two events are mutually exclusive if they cannot happen at the same time.

$$\text{For mutually exclusive events: } P(A \text{ intersection } B) = 0$$

Example: In one coin toss, Head and Tail cannot occur together. In one die roll, getting 2 and getting 5 cannot occur together.

3.3 Independent Events

Two events are independent if the occurrence of one event does not affect the occurrence of the other.

$$\text{For independent events: } P(A \text{ intersection } B) = P(A) \times P(B)$$

Example: The result of the first coin toss does not affect the second coin toss.

3.4 Equally Likely Events

Events are equally likely when every outcome has the same chance of occurring. In a fair die, all six outcomes have probability $1/6$.

Mutually Exclusive	Independent
Cannot occur together.	Can occur together, but one does not affect the other.
$P(A \text{ intersection } B) = 0$	$P(A \text{ intersection } B) = P(A) \times P(B)$
Example: Head and Tail in one toss.	Example: Head in toss 1 and Head in toss 2.

4. Types of Probability

Type	Meaning	Example
Theoretical Probability	Based on mathematical reasoning.	$P(\text{rolling a 3 on a die}) = 1/6$.
Experimental Probability	Based on actual repeated trials.	If 3 occurs 20 times in 100 die rolls, $P(3) = 20/100 = 0.20$.
Subjective Probability	Based on personal judgement or experience.	A cricket expert estimates a 70% chance of winning.

5. Marginal, Joint, and Conditional Probability

5.1 Marginal Probability

Marginal probability is the probability of one event alone. It is not conditioned on any other event.

Example: From a deck of 52 cards, $P(\text{red card}) = 26/52 = 1/2$. $P(\text{four}) = 4/52 = 1/13$.

5.2 Joint Probability

Joint probability is the probability that two or more events happen together. It is the probability of intersection.

$$\text{Joint probability of A and B} = P(A \text{ intersection } B)$$

Example: Probability that a card is both red and four = $2/52 = 1/26$ because only two red fours exist: 4 of hearts and 4 of diamonds.

5.3 Conditional Probability

Conditional probability is the probability of one event occurring when another event has already occurred. It reduces the sample space.

$$P(A | B) = P(A \text{ intersection } B) / P(B)$$

Read $P(A | B)$ as 'probability of A given B'.

Example: Given that a card is red, what is the probability that it is a four? There are 26 red cards and two red fours, so $P(\text{four} | \text{red}) = 2/26 = 1/13$.

6. Addition Rule

The addition rule is used when the question asks for 'A or B', 'at least one', or 'either this or that'.

6.1 If events are mutually exclusive

$$P(A \text{ union } B) = P(A) + P(B)$$

Example: In one die roll, $P(2 \text{ or } 5) = 1/6 + 1/6 = 2/6 = 1/3$.

6.2 If events are not mutually exclusive

$$P(A \text{ union } B) = P(A) + P(B) - P(A \text{ intersection } B)$$

We subtract $P(A \text{ intersection } B)$ because the overlapping part is counted twice.

Example: Find probability that a selected card is a king or a diamond. $P(K) = 4/52$, $P(D) = 13/52$, $P(K \text{ intersection } D) = 1/52$. So $P(K \text{ union } D) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$.

7. Multiplication Rule

The multiplication rule is used when the question asks for 'A and B', 'both events', or 'intersection'.

7.1 Independent Events

$$P(A \text{ intersection } B) = P(A) \times P(B)$$

Example: Probability of Head in first toss and Head in second toss = $1/2 \times 1/2 = 1/4$.

7.2 Dependent Events

$$P(A \text{ intersection } B) = P(A) \times P(B | A)$$

Use this when the first event affects the second event. Common examples include drawing cards or balls without replacement.

Example: A bag has 5 red and 3 blue cards. Probability of drawing two red cards without replacement = $(5/8) \times (4/7) = 20/56 = 5/14$.

8. Odds in Favor and Odds Against

Probability compares favourable outcomes with total outcomes. Odds compare favourable outcomes with unfavourable outcomes.

$$\text{If odds in favor of A are } a:b, \text{ then } P(A) = a / (a + b)$$

$$\text{If odds against A are } b:a, \text{ then } P(A') = b / (a + b)$$

Example: Odds that a person is overweight or obese are 14:11. Then $P(\text{overweight/obese}) = 14/(14+11) = 14/25$. Probability that the person is fit = $1 - 14/25 = 11/25$.

9. Bayes' Theorem

Bayes' Theorem is one of the most important results in probability. It is used to update probability when new evidence is available. It also helps reverse conditional probabilities.

Very often, we know $P(B | A)$, but we need $P(A | B)$. Bayes' theorem allows us to calculate this reverse probability.

$$P(A | B) = [P(A) \times P(B | A)] / P(B)$$

Term	Meaning
$P(A)$	Prior probability of event A before observing evidence B.
$P(B A)$	Probability of evidence B when A is true.
$P(B)$	Total probability of evidence B.
$P(A B)$	Updated probability of A after observing B.

Example idea: A = person has disease, B = test is positive. $P(B | A)$ means probability of positive test if the person has disease. $P(A | B)$ means probability that the person has disease given that the test is positive. Bayes' theorem converts $P(B | A)$ into $P(A | B)$.

10. Total Probability Theorem

The theorem of total probability is used when an event can occur through multiple mutually exclusive cases.

$$P(E) = \text{sum of } P(E | E_i) \times P(E_i)$$

Here, $E_1, E_2, E_3, \dots$ are different cases that do not overlap and together cover the full sample space.

Example: If one urn is chosen randomly from three urns and then one ball is drawn, the probability of drawing a white ball is: $P(W) = P(W|U_1)P(U_1) + P(W|U_2)P(U_2) + P(W|U_3)P(U_3)$.

11. Permutation and Combination

11.1 Permutation

Permutation means arrangement. Use it when order matters.

$$nPr = n! / (n - r)!$$

Example: Selecting rank 1, rank 2, and rank 3 from 5 students uses permutation because order matters.

11.2 Combination

Combination means selection. Use it when order does not matter.

$$nCr = n! / [r! (n - r)!]$$

Example: Selecting any 3 students from 5 students uses combination because order does not matter.

12. Complete Solved Example: Bayes with Red and Blue Cards

Question: A bag contains 5 red and 3 blue cards. Two cards are drawn at random without replacement. Find the probability that the first card is red when the second card is red.

Let A = first card is red. Let B = second card is red.

$$P(A) = 5/8$$

$$P(B | A) = 4/7$$

Now calculate P(B), the total probability that the second card is red. It can happen in two cases:

Case	Probability
First red and second red	$(5/8) \times (4/7) = 20/56$
First blue and second red	$(3/8) \times (5/7) = 15/56$

$$P(B) = 20/56 + 15/56 = 35/56$$

Apply Bayes' theorem:

$$P(A | B) = [P(B | A) \times P(A)] / P(B) = (20/56) / (35/56) = 20/35 = 4/7$$

Final answer: The probability that the first card was red given that the second card is red is $4/7 = 0.571$.

13. How to Choose the Right Formula

Question Keyword	Formula / Method
or	Use addition rule.
and	Use multiplication rule.
given that	Use conditional probability.
at least one	Use addition rule or complement rule.
at most	Add all probabilities up to that number.
without replacement	Usually dependent probability.
with replacement	Usually independent probability.
new evidence	Use Bayes' theorem.
arrangement/order/rank	Use permutation.
selection/team/group	Use combination.

14. Important Exam Questions with Short Answers

Q1. A coin is tossed two times. Find probabilities.

Sample space: {HH, HT, TH, TT}.

Question	Answer
Two heads	1/4
One tail	$2/4 = 1/2$

No head	1/4
At least one head	3/4
At most one head	3/4

Q2. A die is rolled. Given the outcome is odd, find probability that it is prime.

Odd outcomes = {1, 3, 5}. Prime among these = {3, 5}.

$$P(\text{prime} \mid \text{odd}) = 2/3$$

Q3. Two children problem

Assume each child is equally likely to be boy or girl. Sample space: {BB, BG, GB, GG}.

Condition	Required Probability
Both are girls given youngest is girl	Only {BG, GG} possible. Both girls = {GG}. Answer = 1/2.
Both are girls given at least one is girl	Possible {BG, GB, GG}. Both girls = {GG}. Answer = 1/3.

Q4. Selection for at least one post

Ramesh appears for two independent posts A and B. $P(A)=1/6$, $P(B)=1/7$. Find probability of selection in at least one post.

$$P(\text{at least one}) = P(A \text{ union } B) = P(A) + P(B) - P(A)P(B)$$

$$= 1/6 + 1/7 - 1/42 = 7/42 + 6/42 - 1/42 = 12/42 = 2/7$$

Q5. Three students solve a problem

Chances of solving are 1/3, 2/7, and 3/8. Find probability that the problem is solved by at least one.

Use complement: probability problem is not solved by any student.

$$P(\text{not solved}) = (1 - 1/3)(1 - 2/7)(1 - 3/8) = (2/3)(5/7)(5/8) = 50/168 = 25/84$$

$$P(\text{solved}) = 1 - 25/84 = 59/84$$

Q6. Total probability urn problem

There are three urns: U1 has 3 white and 2 black, U2 has 2 white and 3 black, U3 has 4 white and 1 black. Each urn is equally likely. Find probability of drawing a white ball.

$$P(W) = (3/5)(1/3) + (2/5)(1/3) + (4/5)(1/3) = 9/15 = 3/5$$

Q7. Bayes machine defect problem

Machines X, Y, Z produce 1000, 2000, 3000 bolts per day. Defective rates are 1%, 1.5%, and 2%. A bolt is selected and found defective. Find probability it came from X.

Production proportions: $P(X)=1/6$, $P(Y)=2/6$, $P(Z)=3/6$. Defect probabilities: $P(D|X)=0.01$, $P(D|Y)=0.015$, $P(D|Z)=0.02$.

$$P(D) = (0.01)(1/6) + (0.015)(2/6) + (0.02)(3/6) = 1/60$$

$$P(X | D) = [(0.01)(1/6)] / (1/60) = 1/10$$

15. Formula Sheet

Concept	Formula	Use When
Basic probability	$P(E) = n(E)/n(S)$	Simple chance.
Complement	$P(E') = 1 - P(E)$	Opposite event.
Addition rule	$P(A \text{ union } B) = P(A) + P(B) - P(A \text{ intersection } B)$	A or B, overlap possible.
Mutually exclusive	$P(A \text{ intersection } B) = 0$	Events cannot happen together.
Independent	$P(A \text{ intersection } B) = P(A)P(B)$	Events do not affect each other.
Dependent	$P(A \text{ intersection } B) = P(A)P(B A)$	One event affects another.
Conditional	$P(A B) = P(A \text{ intersection } B) / P(B)$	Given that B happened.
Bayes	$P(A B) = P(A)P(B A) / P(B)$	Update probability using evidence.
Total probability	$P(E) = \sum P(E E_i)P(E_i)$	Multiple cases.
Permutation	$nPr = n! / (n-r)!$	Order matters.
Combination	$nCr = n! / [r!(n-r)!]$	Order does not matter.

16. Final Summary

Probability measures uncertainty. Marginal probability is probability of one event alone. Joint probability is probability of two events together. Conditional probability is probability after information is given. Bayes' theorem updates probability using new evidence. The addition rule is used for 'or' questions, and the multiplication rule is used for 'and' questions. The theorem of total probability is used when an event can happen through multiple cases. Permutation is used when order matters, while combination is used when order does not matter.

For exams and interviews, always identify the keywords first: 'or', 'and', 'given that', 'at least', 'without replacement', and 'new evidence'. These words usually tell you which formula to apply.